

Midterm 2

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Current I (A)	Magnetic Field B (μT)
0.50	31.4
1.00	62.8
1.50	94.2
2.00	125.7
2.50	157.1

Table 1: B-field vs. current data.

- What is the direction of the Lorentz force on a negatively charged particle moving in the y -direction near a positive current in the z -direction, located at the origin?
- (a) In which direction is the magnetic moment of a loop of current in the xy -plane, if the current moves counterclockwise when observed from above the xy -plane? (b) If an external, uniform magnetic field is activated and points in the x -direction, what is the direction of the torque vector on the loop? (c) What will be the final orientation of the loop? *Qualitatively, the alignment of molecular and atomic magnetic moments is how magnetization arises in solids.*
- A solenoid of length 20 cm is connected to a DC power supply. A magnetic field probe is placed at the center of the solenoid to measure the B-field. The data are shown in Tab. 1. (a) How many turns are in the solenoid? (b) What B-field is generated with 5.0 A of current?
- (a) An MRI technician moves his hand from a region of very low magnetic field strength into an MRI scanner's 2.00 T field with his fingers pointing in the direction of the field. Find the average emf induced in his wedding ring, given its diameter is 2.20 cm and assuming it takes 0.250 s to move it into the field. (b) What current is induced in the ring if its resistance is $0.0100\ \Omega$? (c) Calculate the average power dissipated in the ring.
- (a) A bicycle generator rotates at 1875 rad/s, producing an 18.0 V peak emf. It has a 1.00 by 3.00 cm rectangular coil in a 0.640 T field. How many turns are in the coil? (b) What is the period of the AC output of the generator?
- Camera flashes charge a capacitor to high voltage by switching the current through an inductor on and off rapidly. In what time must the 0.100 A current through a 2.00 mH inductor be switched on or off to induce a 500 V emf?
- A large research solenoid has an inductance of 25.0 H. (a) What induced emf opposes shutting it off when 100 A of current through it is switched off in 80.0 ms? (b) How much energy is stored in the inductor at full current? (c) At what rate in watts must energy be dissipated to switch the current off in 80.0 ms? (d) Design a solenoid that has an inductance of 25.0 H by choosing the number of turns N , the cross-sectional area A , and the number of turns l . (d) What is the reactance of the inductor at 10 kHz?
- Your RL circuit has a characteristic time constant of 20.0 ns, and a resistance of 5.00 M Ω . (a) What is the inductance of the circuit? (b) What resistance would give you a 1.00 ns time constant, perhaps needed for quick response in an oscilloscope? (c) What percentage of the final current I_0 flows 3.0 ns after deactivation?
- Consider Fig. 1. Assume the bottom wire is grounded at 0V. (a) Using Kirchhoff's loop rule on the left side of the circuit, write an equation that describes changes in potential. (b) Repeat exercise (a) for the right loop. (c) Solve parts (a) and (b) for v_{in} and v_{out} , respectively, and calculate $v_{\text{out}}/v_{\text{in}}$ in terms of R and C . (d) Is this a low-pass (LP) or high-pass (HP) filter? Argue for LP or HP by *taking the limit* of frequency.
- (a) What is the resonance frequency, f_0 , of an RLC circuit with $R = 0.1\ \text{k}\Omega$, $C = 1\ \mu\text{F}$, and

$L = 10 \text{ mH}$? (b) What is the total impedance of this circuit at $f = 0.1f_0$ and $f = 10f_0$?

θ_1 (deg)	θ_2 (deg)
20.0	13.2
30.0	19.5
40.0	25.0
50.0	30.0
60.0	35.0

Table 2: Incident and refracted angles.

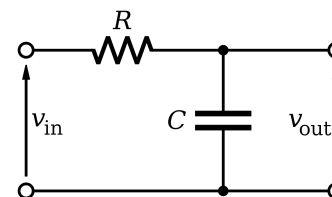


Figure 1: This RC circuit can act as a low-pass filter.

11. For an RLC circuit with that values of R , L , and C from the previous problem, an AC source voltage characterized by $V_{\text{rms}} = 120 \text{ V}$, (a) what is I_{rms} at $3f_0$? (b) What is P_{rms} at this frequency? *Hint: calculate the power factor, the cosine of the phase angle between resistance and reactance.*
12. (a) What is the B-field generated 1 cm laterally from a capacitor in an RC circuit that charges from 0 to 200 nC in $2 \mu\text{s}$? (b) What *current* is responsible for this B-field, since the capacitor is not passing DC current like a wire?
13. (a) An aircraft receives a pulse reflected from glacial ice below it $10 \mu\text{s}$ after transmission. What is the displacement between aircraft and ice? (b) What fraction of the power is reflected from the ice, assuming standard indexes of refraction for air and ice at RF wavelengths?
14. (a) A microwave generates 1 kW of power, projected onto a 10 cm-squared area at a distance of 1 m. What is the intensity (power per unit area)? (b) If the intensity drops as the distance squared, what is the intensity 2 meters from the microwave?
15. (a) Show¹ that $\theta_3 = \theta_1$ in Fig. 2, if $n_1 = n_3$. (b) A doctor examines the ear of a child using a 12.0 cm focal length eyepiece 8.0 cm from the ear. What is the magnification?
16. A beam of visible light traveling through air strikes the surface of a transparent material. The angle θ_1 (measured from the normal) and the refracted angle θ_2 are measured and shown in Tab. 2. (a) What is the average and standard deviation of the measured index of refraction? (b) Plot $\sin \theta_2$ versus $\sin \theta_1$ to show that the data are linearly correlated. Confirm these results with the appropriate PhET simulation and plot both the simulation data and the data from Tab. 2 on the same graph.
17. Suppose x-rays have a cross section of 200 barns at 10 keV in a lead vest protecting a medical patient. If the vest is 1 cm thick, what fraction of the x-rays pass through it?

18. (a) What is the radioactive dose in rads if 250 mJ of total energy is deposited in the body of a 60 kg adult? (b) What instead would be the dose if the radiation was concentrated in just 2.0 kg of tissue? (c) Assuming the RBE is 1 (x-rays or γ rays), what is the dose from part (b) in Sv? (d) Does this dose represent a serious health risk?

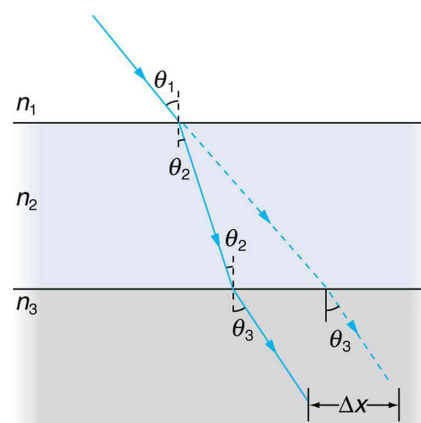


Figure 2: A light ray is in a medium with n_1 , then enters a medium with n_2 , then a medium with n_3 .

¹This proof is the reason why the center of our lenses (like those in our glasses) do not change ray directions. If our glasses changed the apparent location of objects, they would be useless.